ACCELERATORS For CAMERAS And AUDIO APPLICATIONS

The immense pace at which the technology is witnessing advancements is also pushing both the software and hardware developers to implement these changes to remain in sync with time. Let us find out how that is shaping the future of cameras, audio applications, and IoT devices

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Today we have a great variety of multimedia applications, including the applications for audio processing, speech recognition, automatic machine translation, text-to-speech, speaker recognition, and authentication. We also have image classification, object (face, person) detection, and semantic segmentation for imaging prospects.

These images require a lot of computes in order to enhance the image quality. This can be done with a wide array of products, such as mobile devices, automotives, wearables, and the Internet of Things (IoT). A lot of this can also be done using the edge devices but often we tend to offload a lot of the computing into cloud and data centres.

Spectrum of compute engines for multimedia

There are many options available at hand today. These are various compute engines (see Fig. 1) where you have varying degrees of energy efficiency and flexibility.

The left most side in Fig. 1 has the lowest energy efficiency but with higher flexibility. There are central processing units (CPUs), graphics processing units (GPUs), general neural processing units (NPUs), specialised NPUs that can take on various types of nn workloads and then you can also go for higher level of energy efficiency with specialised engines.

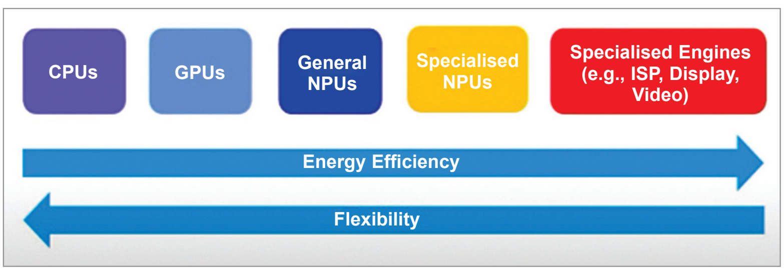

Fig. 1: Spectrum of compute engines

The right most side engine can process a significantly larger amount of data with the highest efficiency. It comes at a trade-off for least amount of flexibility.

Specialised engines for high efficiency

The following are some of the examples for specialised engines.

Let us take a look at image signal processor (ISP) as a starting point for cameras. It takes in the raw input from the sensor with the data rate, for instance 12MP @ 60 frames per second (fps) + 108MP @ 15 fps. A lot of times the 15fps images are fused to form a high-quality image. Once the raw data is taken it generates the final enhanced YUV images. You can perform image restoration along the way such as white balancing, lens shading correction, colour correction, gamma correction, and tone mapping. It can either be local or global. There are also options for denoising, sharpening, and scaling.

Then there is video encoder and decoder (for instance, 1080P @ 30fps or 4k @ 60fps). A lot of the time they allow you to capture videos and encode them in the compressed stream or you can playback the compressed video and show it for display. They typically support multiple formats as shown and they are also highly versatile in a sense that they can do both decoding as well as encoding and support multiple formats.

The display engine scales and mixes multiple sources and then performs some quality enhancements, such as toe mapping and various other corrections. It also does colour space convergence to render the final